

Mini Project:

Material/Printer Tuning, and Design

Due: Wednesday, February 25 – 5PM

Weight: 10%

Project Overview

In this project, you will tune a 3D printer for a specific filament and color, document how different settings affect print quality, and then design and print a simple two-part assembly informed by what you have learned.

This project is intended to help with understanding how to select print settings and prepare you for the final design project, where you will create your own unique design.

Learning Objectives

- Calibrate a 3D printer for a specific filament
- Identify common print issues and adjust settings accordingly
- Make basic design decisions based on material behavior
- Explain why a design/settings work(or doesn't)
- Document and communicate your design process clearly

Project Tasks:

Filament Calibration & Testing

Using one of the four filament colors available for the project and the provided calibration models, complete the following testing/tuning protocols

- Temperature tower
- First-layer calibration
- Retraction test
- Overhang and/or bridging test
- Calibration Cube

You do not need to design the calibration parts—STLs will be provided. **You are expected to document all of your prints, including improvements or any failed prints**

Each student is permitted to use **up to 200 grams of material** during this part of the project.

Settings Selection

- Based on your testing results select a set of settings that best suit your design/application.

Design & Final Print

Using what you learned about your filament, design two simple objects that fit/connect together within the following design constraints:

- Two, **unique**, fully printed parts designed by you (inspiration can be found from existing designs but no downloaded parts are permitted)
- Each part must fit within a 100 × 100 × 100 mm volume
- **The maximum permitted amount of material is 150 grams per part**
- Parts must be printed with the color and parameters you chose.
- The design must include each of the following:
 - An overhang (on at least one of the parts, can be on both)
 - A fit feature that connects the two parts (slot, clip, peg, bolt pattern etc.)
 - A surface finish or orientation decision (applicable to both parts)

Designs could include: small containers, clips, stands, or simple tools/jigs, etc...

Consult with the course instructor if you anticipate needing to use fasteners/mechanical hardware for your design

Required Deliverable (1 PDF document)

- Images of all calibration prints and brief commentary on the printing results
- A table of your chosen print settings (such as layer height, wall count, top/bottom layers, speed, bed and nozzle temps, infill type/%, support vs no support, bed adhesion, etc...)
- **A justification for why the chosen settings are appropriate for the design that you will be printing (consider the functionality/strength requirements of your design) Clear reasoning matters more than perfect prints right now.**
- Screen shots of all sketches and features (similar to assignment 3/4)
- Photos of your final printed objects assembled together (also presented in class)
- A Short 2-3 paragraph commentary on what you learned through this process, challenges faced, what worked/didn't work with your design and how you intend to apply the learnings to your final project

All images/commentary should be compiled into a single neatly organized PDF document

Rubric

Calibration Prints & Documentation of Results – 35%

Settings Justification – 25%

Final Design & Print (including CAD screenshots) – 25%

Reflection – 10%

Organization/Neatness – 5%

Note: Print quality/success alone will not determine your grade. Understanding why something worked or didn't work is more important.